

Reviewer Pack — Minimal Tropical Hodge Index Correlation with DPLL Search Tr...

Entry 8c67a9507f77 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Tropical Hodge Index Correlation with DPLL Search Tree Diameter

Entry ID: 8c67a9507f77 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-01 15:38:30 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry **Field B** (complexity object): DPLL search trees

Statement:

For every CNF φ , the minimal tropical Hodge index ($h(\varphi)$) of its associated matroid is linearly correlated with its DPLL search tree diameter $d(\varphi)$, such that $h(\varphi) = \Omega(d(\varphi))$. Furthermore, for any fixed $\varepsilon > 0$, there exists a constant c_ε such that for all CNFs φ with n variables, $|h(\varphi)| \geq c_\varepsilon * d(\varphi)$.

Rationale (proposer's reasoning):

Tropical geometry provides a framework to study tropical varieties, which can be used to represent matroids. The Hodge index of these varieties has been previously applied in algebraic combinatorics but rarely in complexity theory. If the conjecture holds, it would provide a new perspective on the structure of DPLL search trees by linking them with geometric properties.

Taxonomy category: TROPICAL_GEOMETRY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 2032434feebca6c

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a CNF φ , if $|h(\varphi)| \geq c_\varepsilon * d(\varphi)$ for all seeds with n variables ($n \leq 40$) and for at least 95% of the seeds, then $h(\varphi) = \Omega(d(\varphi))$ is supported.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Minimal tropical Hodge index AND DPLL search tree diameter - Tropical geometry AND DPLL search trees AND correlation - CNF matroid AND tropical Hodge index AND DPLL tree diameter $\geq$ constant ε

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=9.0s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        clauses = []
        for _ in range(2**n):
            clause = [random.choice([-1, 1]) * (i + 1) for i in range(n)]
            if any(c != 0 for c in clause):
                clauses.append(clause)
        return clauses

    def dpll(cnf, assignment={}):
        if not cnf:
            return True
        literal = next((l for l in range(1, len(cnf[0]) + 1) if l not in assignment and -l not in
            if literal is None:
                return False

        def propagate(lit):
            new_cnf = []
            for clause in cnf:
                if any(abs(x) == abs(lit) for x in clause):
                    continue
                new_clause = [x for x in clause if x != -lit]
                if not new_clause:
                    return None
                new_cnf.append(new_clause)
            return new_cnf

        def backtrack():
```

```

        nonlocal assignment
        last_lit = list(assignment.keys())[-1]
        del assignment[last_lit]
        for lit in range(last_lit + 1, len(cnf[0]) + 1):
            if lit not in assignment and -lit not in assignment:
                assignment[lit] = True
                new_cnf = propagate(lit)
                if new_cnf is not None:
                    if dpll(new_cnf, assignment):
                        return True
                del assignment[lit]
        return False

assignment[literal] = True
new_cnf = propagate(literal)
if new_cnf is not None:
    if dpll(new_cnf, assignment):
        return True
del assignment[literal]

assignment[-literal] = True
new_cnf = propagate(-literal)
if new_cnf is not None:
    if dpll(new_cnf, assignment):
        return True
del assignment[-literal]

return backtrack()

def hodge_index(cnf):
    # Placeholder for actual Hodge index computation
    return random.random() # Simplified for testing

n = random.choice([5, 10, 15, 20, 30, 40])
cnf = generate_cnf(n)
h = hodge_index(cnf)
d = len(dpll(cnf))

return {
    "metric_name": "Hodge Index vs DPLL Diameter",
    "metric_value": abs(h) / (d + 1), # Avoid division by zero
    "instances_tested": 1,
    "n_max": n,
    "conjecture_holds": False,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

```

```

if all(r["conjecture_holds"] for r in results):
    mean_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = 1.0
    print(f"RESULT: SUPPORTED mean={mean_value} std=0 support_fraction={support_fraction}")
else:
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
    print(f"RESULT: FALSIFIED counterexample=' ' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b00917d6.py", line 101, in <module>
    result = run_trial(seed)
              ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b00917d6.py", line 84, in run_trial
    d = len(dpll(cnf))
          ~~~~~^
TypeError: object of type 'bool' has no len()

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed during execution, which prevents us from verifying the conjecture's support conditions. | next: Re-examine the test code for potential bugs and ensure it can run to completion without errors.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14822
2	preregistration	ollama_remote	glm4:latest	0	0	9543
3	novelty	ollama_remote	glm4:latest	0	0	9284
4	novelty	ollama_remote	glm4:latest	0	0	9901
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16506
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	29454
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17723
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	23239

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
9	judge	ollama_remote	glm4:latest	0	0	11049

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 141523 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/8c67a9507f77.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/8c67a9507f77.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/8c67a9507f77.tar.gz (if generated)